

Cloud Computing Experiment No 4

Name: Parth Vijay Gulhane

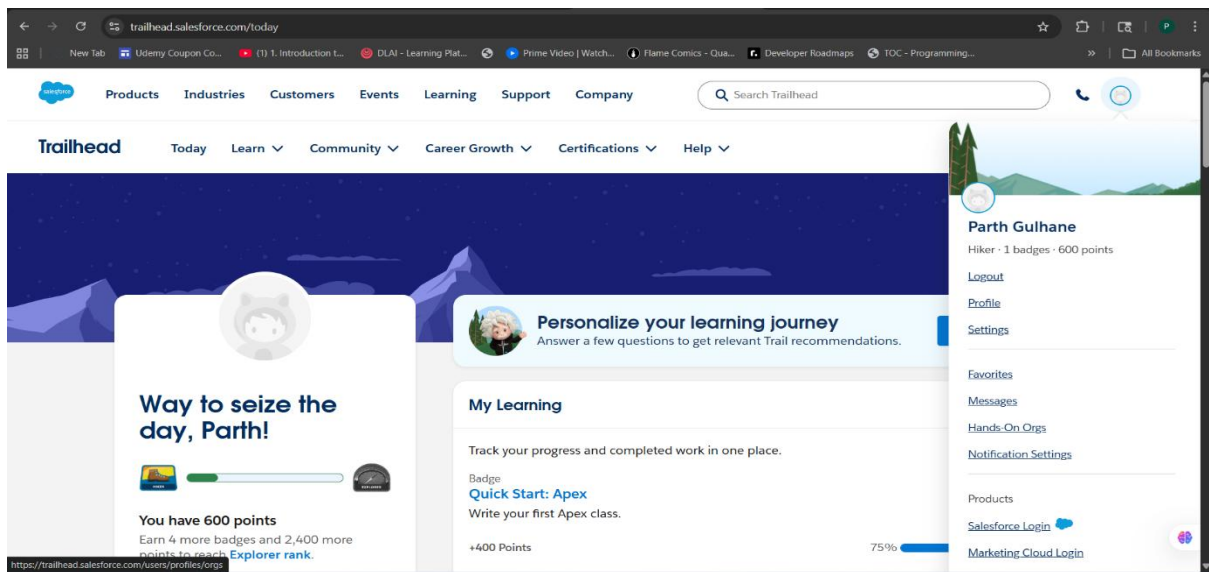
Roll No: 23CO043

Class: TE Comp A (Batch B)

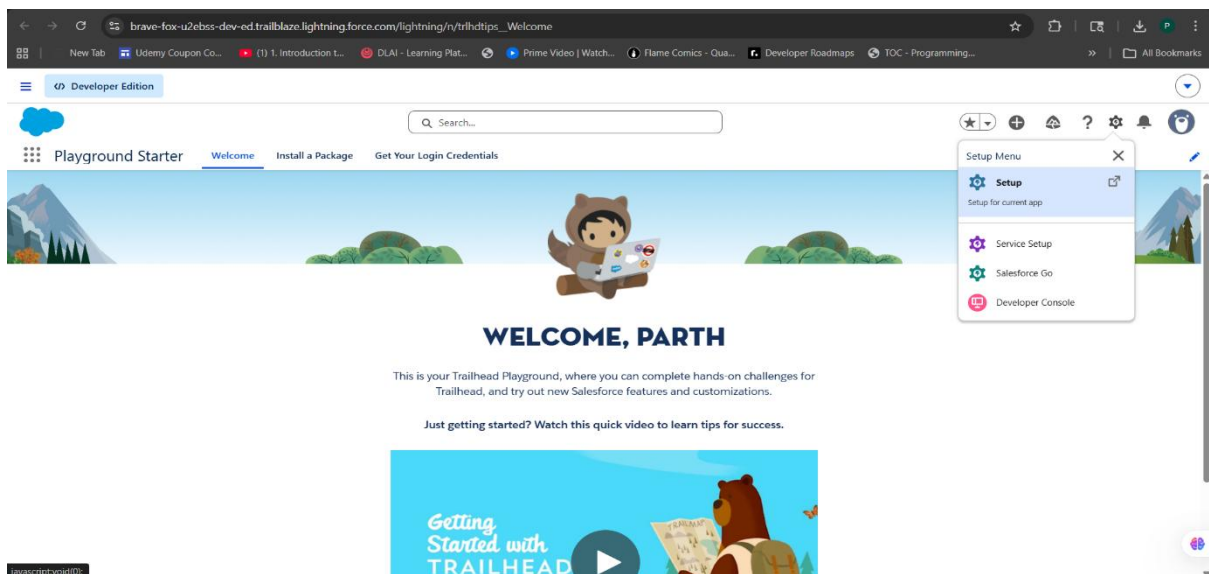
Aim: Design and develop custom Application (Mini Project) using Salesforce Cloud.

Step by Step Procedure:

Step 1: Signup/Login to Trailhead Salesforce

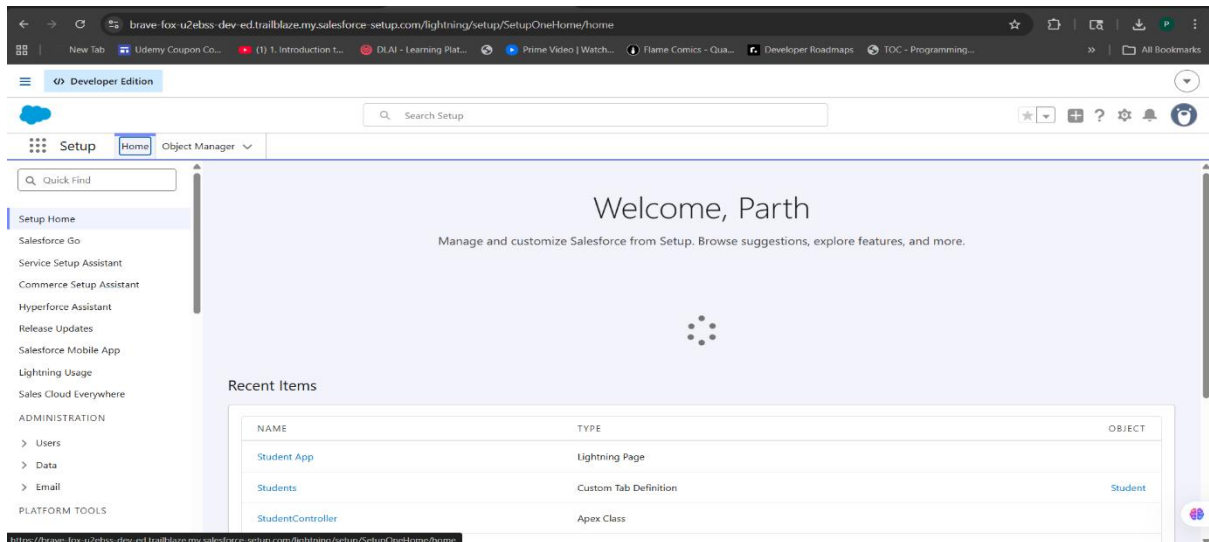


Step 2: Click on Profile Icon (Top Right) then Click Hands-on Orgs and launch the Playground



Step 3:

1. Look at top right  icon
2. Click → **Setup**



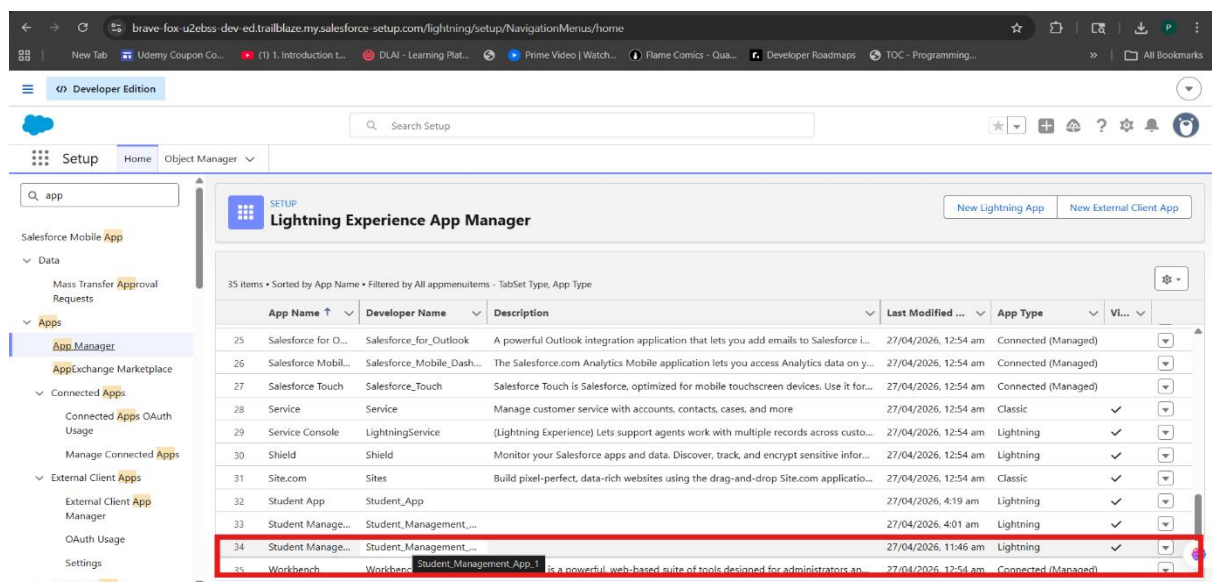
Step 4:

In Left Search Bar search **App Manager** open it and click on **New Lightning App**

Step 5: Fill details

- App Name -> Student Management App 1

Leave everything same and click **Save & Finish**



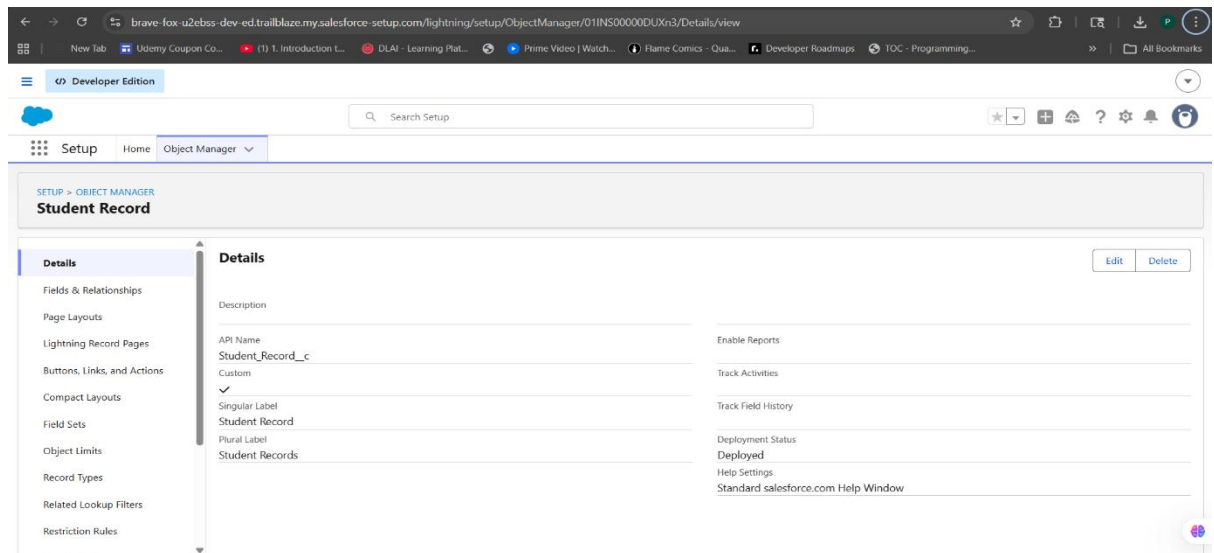
Step 6: Create object

Go to Setup → **Object Manager** → Create → Custom Object

Label -> Student Record

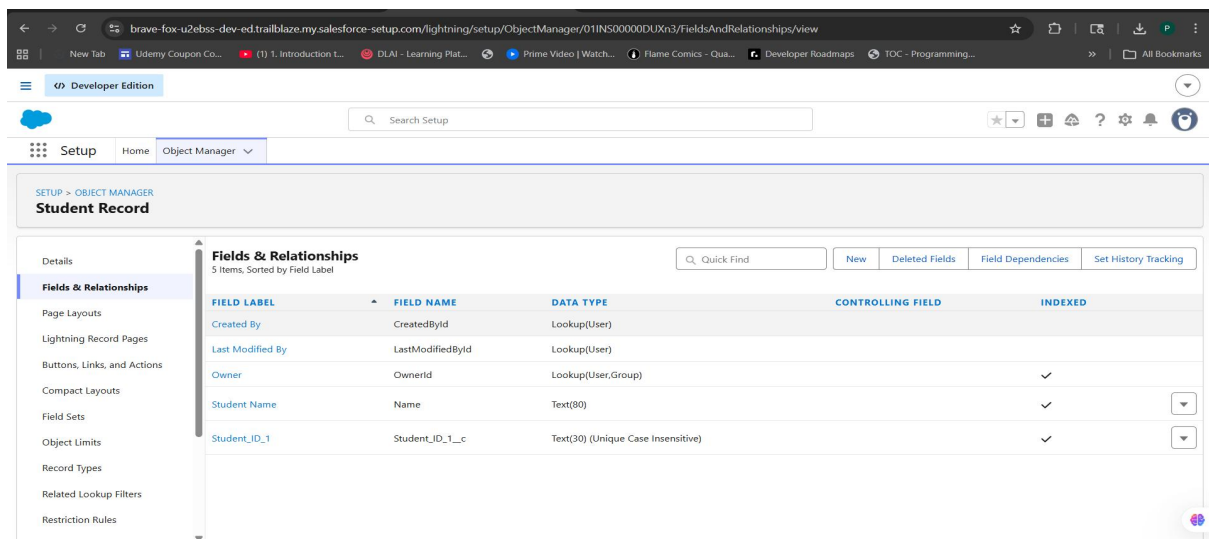
Plural -> Student Records

Record Name -> Student Name



Step 7: Add Fields

- Student ID
 - Type -> text
 - Label -> Student_ID_1
 - Unique



Step 8: Create Object Subject

- Label -> Subject Record
- Record Name -> Subject Name

Step 9: Add Fields

- Marks
 - Type -> Number
 - Label -> Marks_1
 - Length -> 3

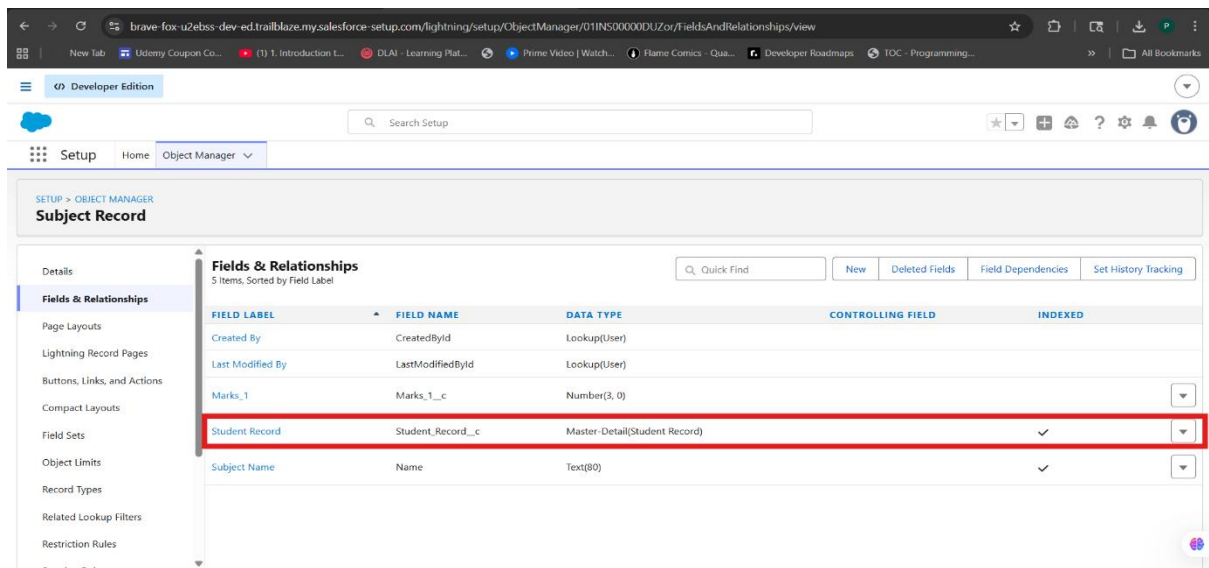
Step 10: Create Relationship

Go to Object Manager -> Subject Record -> Fields & Relationship -> New

Select Master-Detail relationship

Fill

- Related To → Student Record
- Field Label → Student



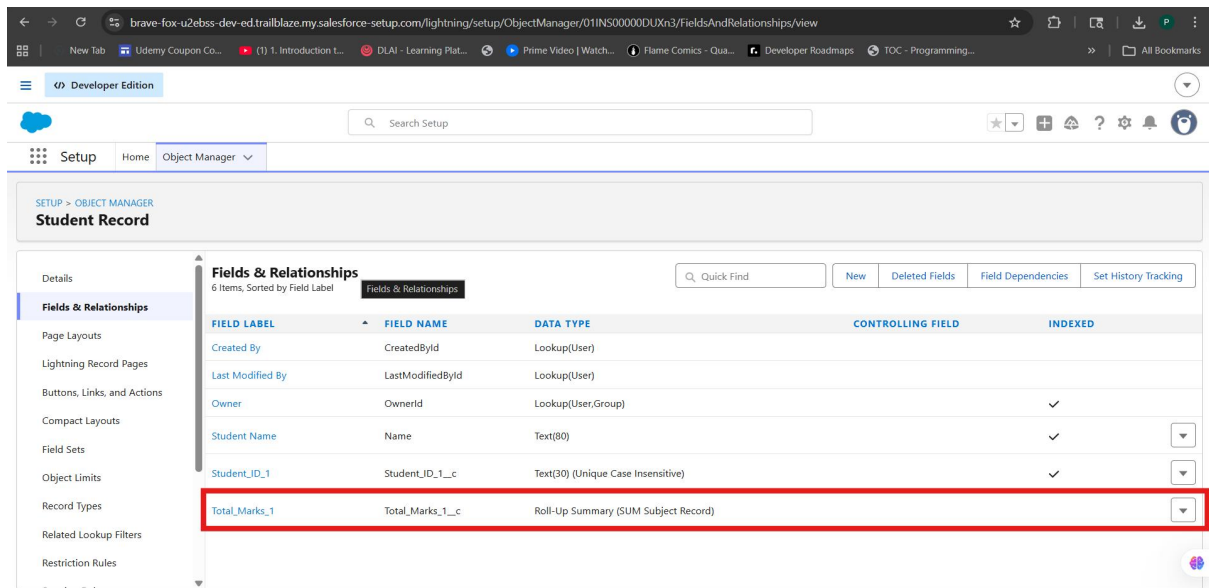
Step 11: Create Total Marks

Go to Object Manager -> Student Record

Fields & Relationship -> New -> Roll-Up Summary

Fill

Field	Value
Label	Total_Marks_1
Object	Subject Record
Type	SUM
Field	Marks_1



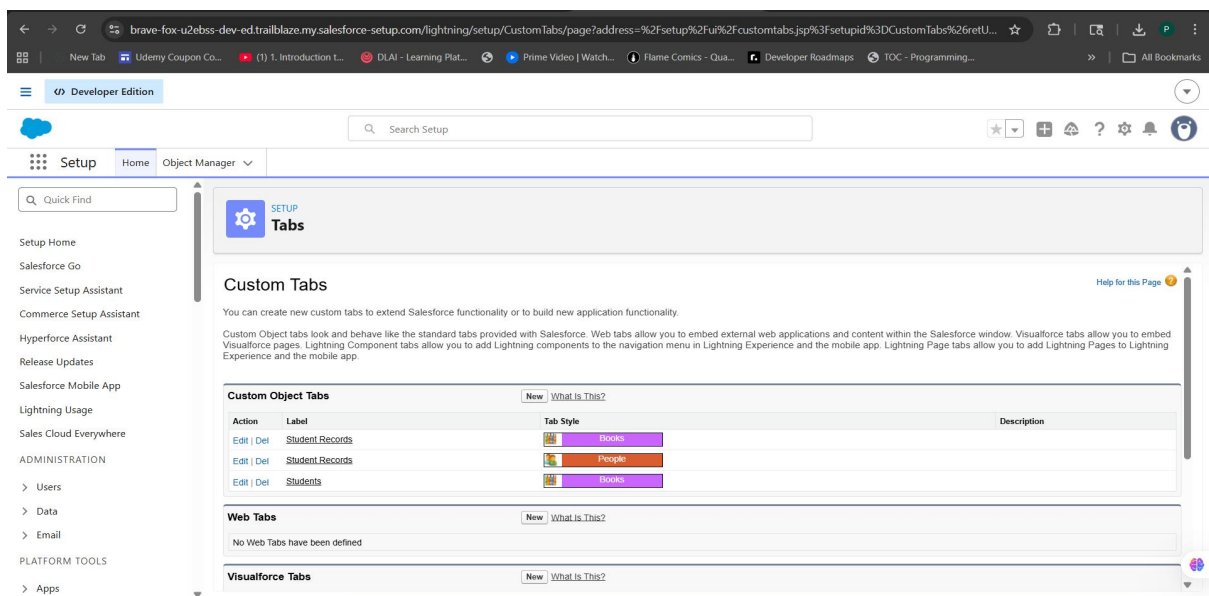
Step 12: Create tabs

Go to Setup -> Search Tabs

Under Custom Object tabs -> new

Create

- Student Record tab
- Subject record tab



Step 13: Add to App

App Manager → Edit **Student Management App 1**

Add:

- Student Records

- Subject Records

Save

Step 14:

Object Manager → Student Record → Page Layouts

Open → **Student Record Layout**

From TOP panel (Fields):

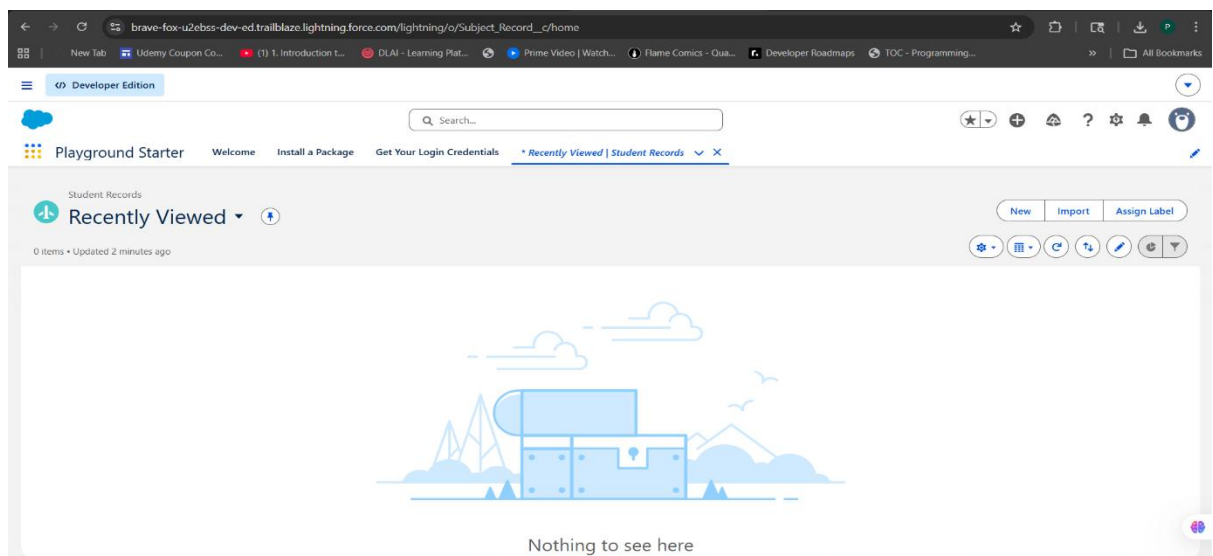
Drag:

- Student_ID_1
- Total_Marks_1

Click **Related Lists**

- Drag:
✓ Subject Records

Step 15: Launch App



Step 16:

Create Student

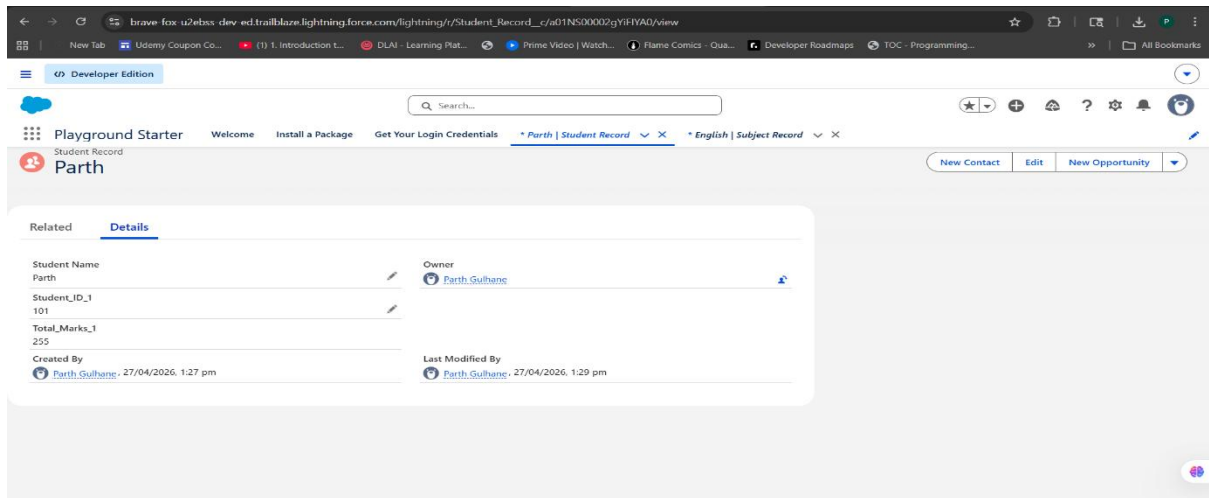
- Name -> Parth
- Student_ID_1 -> 101

Add Subject

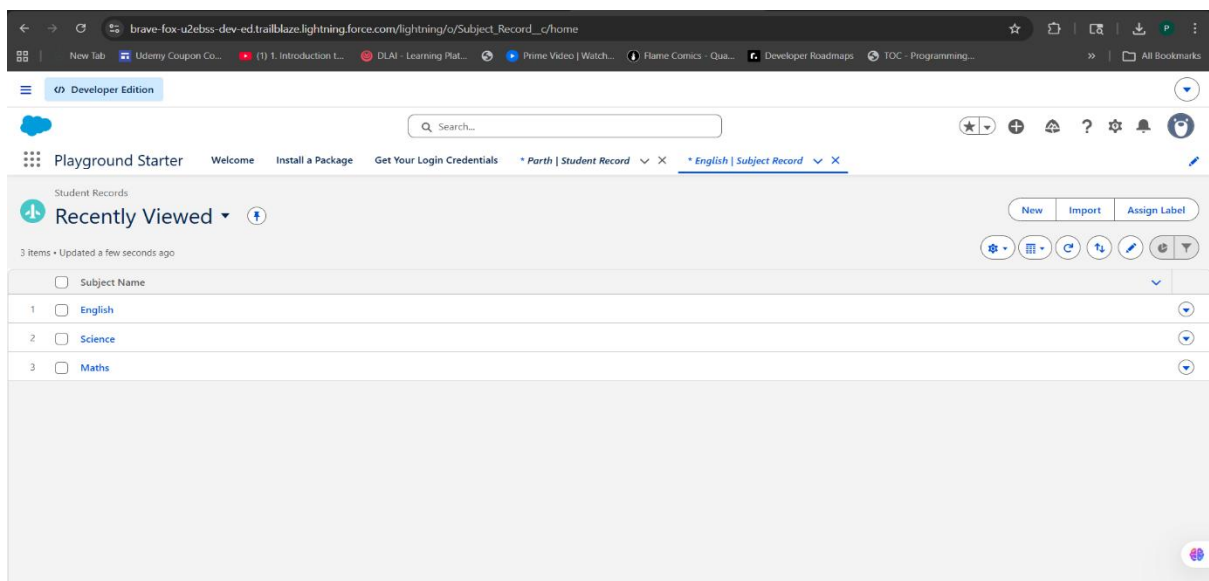
- Maths 85
- Science 90

- English 80

And relate them to student Parth



Output : Total_Marks_1 -> 255



Step 17: Add APEX CODE (Custom Function)

Go to Setup → Apex Classes → New

And paste write this code

```
public class StudentMarksController_Final { public static void calculateTotal(Id studentId) {  
  
    List<Subject_Record__c> subs = [  
  
        SELECT Marks_1__c FROM Subject_Record__c  
  
        WHERE Student__c = :studentId];
```

```

Integer total = 0;

for(Subject_Record__c s : subs){

    total += s.Marks_1__c;}

Student_Record__c stu = [

    SELECT Id, Total_Marks_1__c

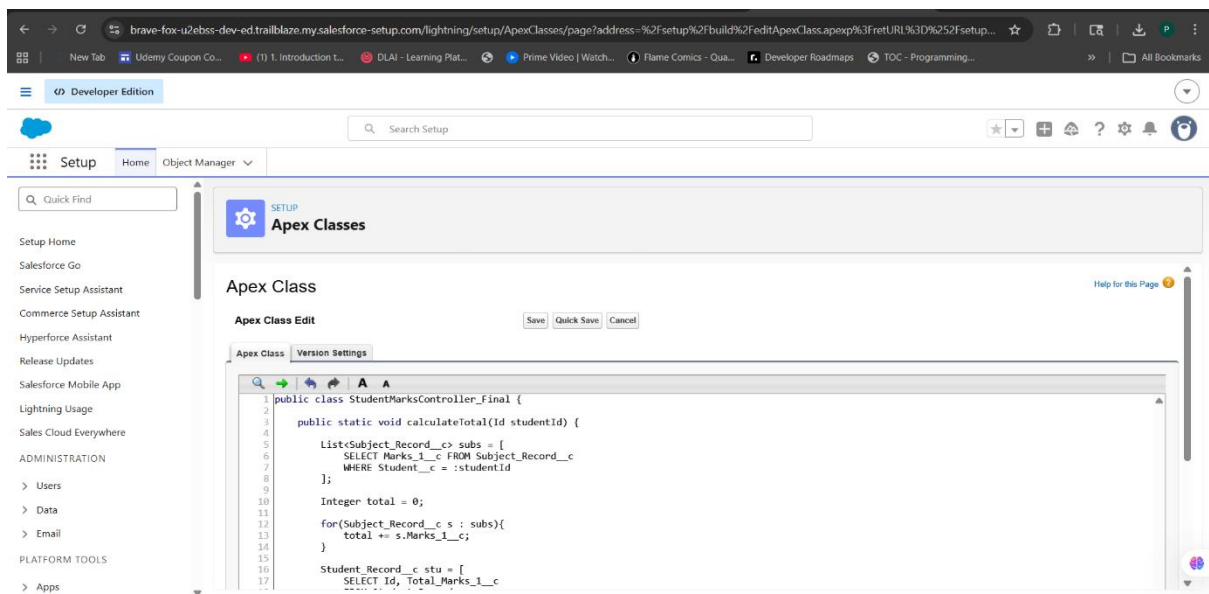
    FROM Student_Record__c

    WHERE Id = :studentId];

stu.Total_Marks_1__c = total;

update stu;}}

```



Step 18: Validation Rule

Open Setup -> Object Manager -> Subject Record -> Validation Rules -> Click on New

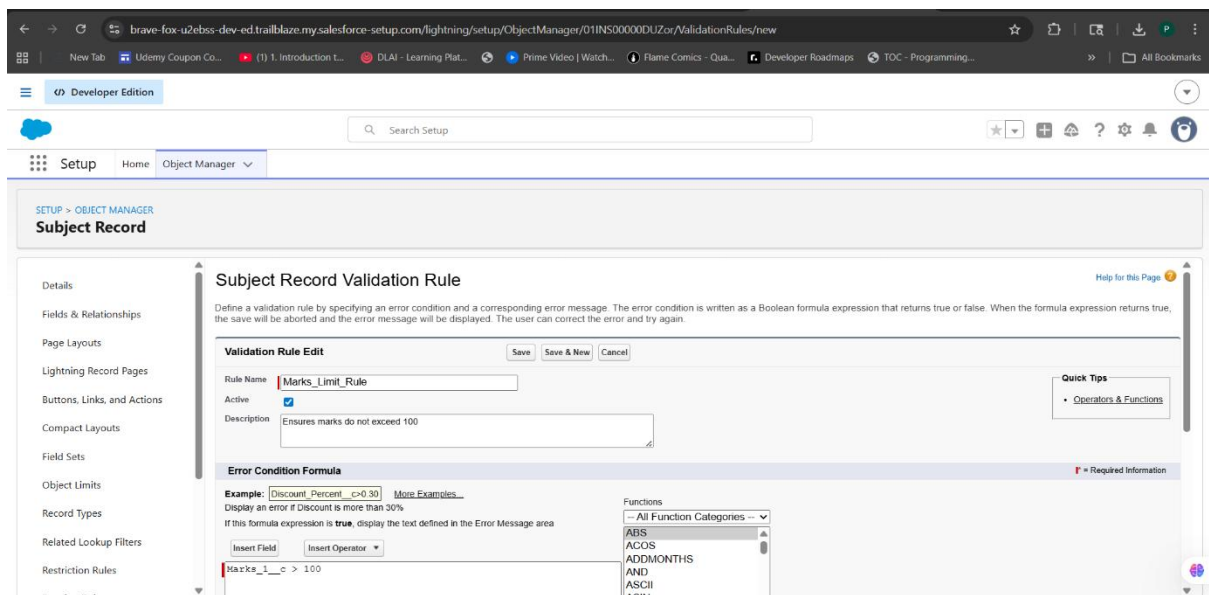
Rule Name -> Marks_Limit_rule

Description -> Ensures marks do not exceed 100

Formula -> Marks_1__c > 100

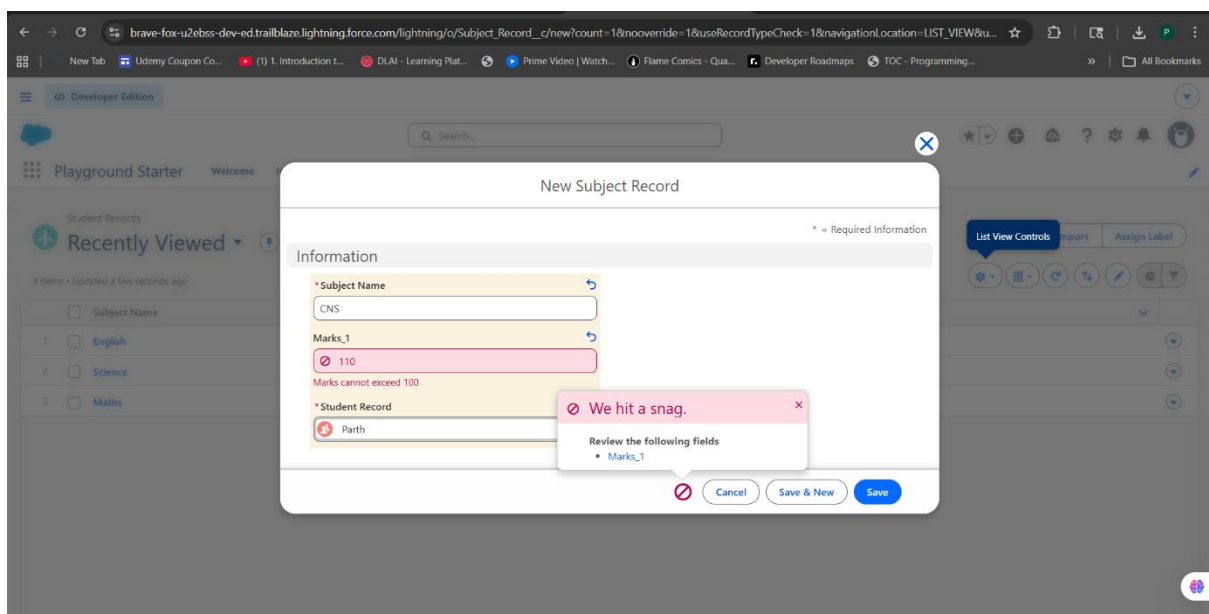
Error Message -> Marks cannot exceed 100

Select Field: Marks_1 and save

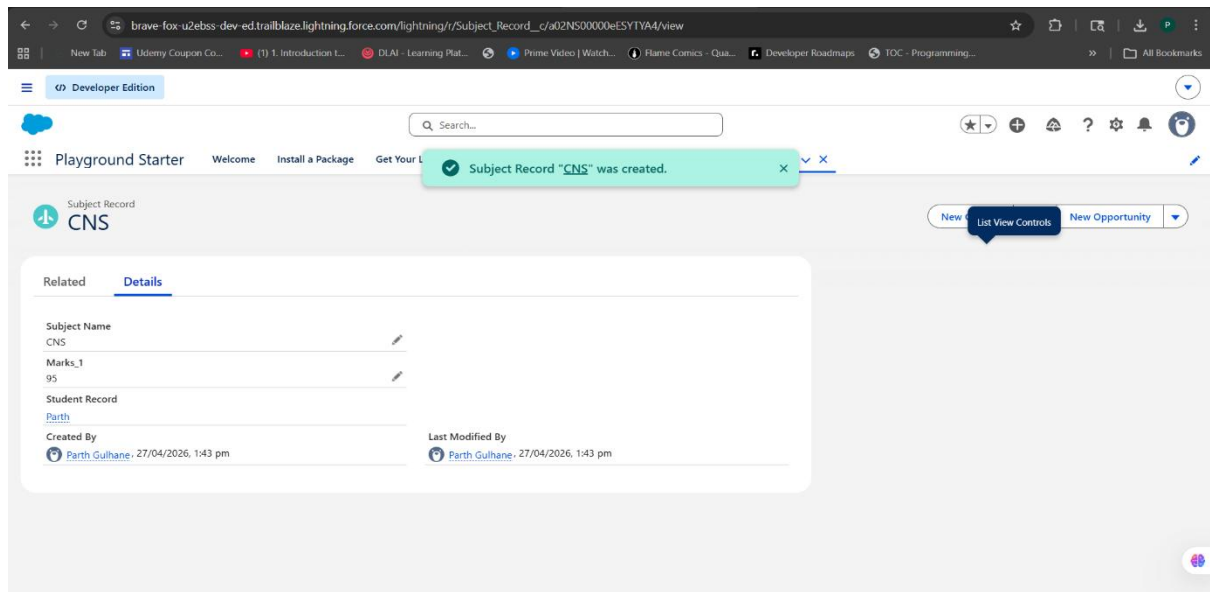


Step 19: Test Your validation

1. Go to your app
2. Open Student
3. Add Subject
4. Marks = 110 -> Error Shown



5. Marks = 95 -> Subject Stored



Conclusion:

The Student Marks Management Application was successfully developed using Salesforce Trailhead. Custom objects and a Master-Detail relationship were used to manage student and subject data, while a Roll-Up Summary field automatically calculated total marks. Validation rules ensured data accuracy, demonstrating how Salesforce can be used to build efficient, data-driven applications.